
LINGUISTICS OF AMERICAN SIGN LANGUAGE

An Introduction

Fifth Edition

CLAYTON VALLI
CEIL LUCAS
KRISTIN J. MULROONEY
MIAKO VILLANUEVA

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UNIT 1

Defining Language

WHAT IS A LANGUAGE?

Human beings communicate and share information through language. All languages, whether signed or spoken, are based on specific rules that the users of the language know and follow. Languages are *rule-governed communication systems*. The scientific study of language is called *linguistics* and the people who study language are called *linguists*. Linguists are interested in discovering and describing the rules that govern language. Without these rules, people would not have a useful communication system — in other words, they would not be able to communicate their thoughts and ideas to one another.

People and animals can communicate through different kinds of communication systems. Some of these systems also have rules. For example, Morse code, semaphore (the flag system used in the United States Navy), traffic signals, and symbols used in public places, as well as the communication systems used by bees, birds, dolphins, and nonhuman primates, all have specific rules that determine meaning. Both Morse code and semaphore use symbols to represent letters of the alphabet, numbers, and, in the case of Morse code, punctuation. In essence, these systems are “codes for codes,” or codes for a writing system. Language shares some features with these other communication systems, but it is distinguished by a number of other features. In the following section we will explore how language is similar to but different from other communication systems.

Features Shared by Language and Other Communication Systems

Systems Use Symbols. Language and other communication systems are composed of symbols that their users manipulate to produce meaning. Different kinds of symbol systems exist to facilitate linguistic communication. English has a writing system that uses symbols to represent sounds or combinations of sounds. For example, the written letter *a* is a symbol for one sound in the English word *cat*, and the combination of written letters is a symbol for an entity in the real world; *cat* is a symbol for a small mammal having a tail, whiskers, etc. The spoken English word is a symbol, part of a different symbolic system separate from the written system,

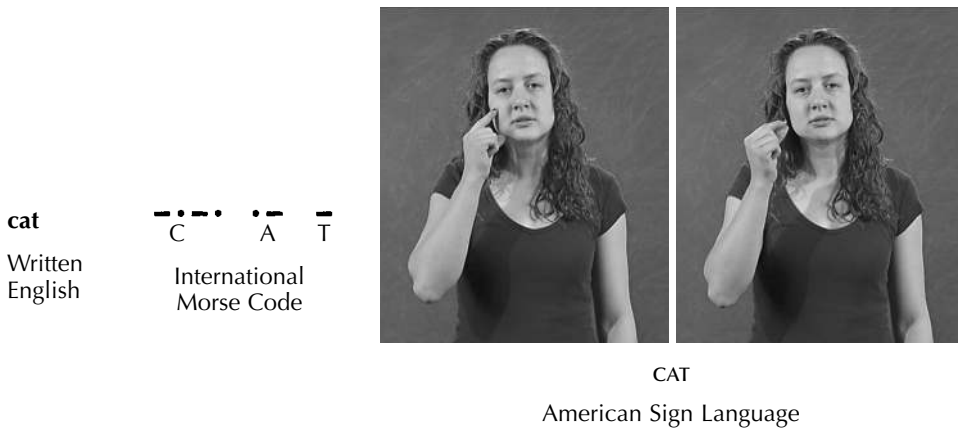


Figure 1. Symbolic representation of “cat.”

Note: This is one of several variants of the sign CAT.

though not unrelated to it. The American Sign Language (ASL) sign CAT is also a linguistic symbol (see Figure 1). (Small capital letters are used for the English word that corresponds to the ASL sign. This is called a *gloss* and will be discussed at length in part 2.)

Symbols Are Organized and Used Systematically. Languages are rule-governed systems, and it is the job of linguists to discover what the rules are and how the system works, a job that is not always easy. The rule-governed nature of ASL can be explained, in part, by examining the conditions on the formation of ASL signs. These conditions were first described by Robbin Battison (1978) as a result of his observations about the structure of ASL signs (see pp. 193–212). Battison proposed that sign formations were based on two conditions, which he called the Symmetry Condition and the Dominance Condition. The Symmetry Condition states that in a two-handed sign, if both hands move, then they will have the same handshape and type of movement. This is illustrated by the signs DRAMA and MAYBE. The Dominance Condition states that in a two-handed sign, if each hand has a different handshape, then only the active hand can move; the passive, or weak, hand serves as a base and does not move. (For right-handed signers, the right hand is the active or dominant hand, while the left hand is the passive or weak hand. The opposite is true for left-handed signers.) The Dominance Condition is illustrated by the signs WORD and MONEY.

When a two-handed sign has different handshapes, Battison reported that the passive hand tends to be one of seven basic handshapes—B, A, S, O, C, 1 (or G), or 5 (see Figure 2). From this information, it is clear that sign structure is not random. Signs can be grouped into different classes; for example, signs like DRAMA and MAYBE are systematically different from signs like WORD and MONEY.



PREACH
(handshape has
no meaning)



NINE-MONTHS
(handshape has
specific meaning)

Figure 7. A comparison of signs in which handshape has no meaning and specific meaning.

types of bee dances, but the units that make up the dances cannot be recombined to make new dances; some research evidence shows the existence of smaller parts in birdsong and primate calls; the design on a semaphore flag is constant and discrete, as is the number of beeps in a Morse code signal. However, a fascinating fact about language is that the symbols of which it is composed can be broken down into smaller parts. In linguistics, this fact is called *duality of patterning*. In language, meaningless units are combined to form arbitrary symbols, and these symbols can in turn be recombined to create larger meaningful units. This point can be illustrated by comparing the signs LOUSY, AWKWARD, and PREACH with the signs THREE-WEEKS, THREE-DOLLARS, and NINE-MONTHS. All six signs have internal structure: they each have a handshape, a location, a palm orientation, and movement. In ASL, the separate parts of some of the signs also have independent meaning. In the signs THREE-WEEKS, THREE-DOLLARS, and NINE-MONTHS, the handshape has specific meaning, such that the sign THREE-WEEKS is different from TWO-WEEKS, the sign THREE-DOLLARS is different from FIVE-DOLLARS, and NINE-MONTHS is different from SIX-MONTHS.

In other signs, all the parts together have one meaning. For example, in LOUSY, AWKWARD, and PREACH, the handshape has no meaning; all of the parts of the sign function together as a unit (see Figure 7).

More than One Meaning Can Be Conveyed by a Symbol or a Group of Symbols. In other communication systems, each symbol or group of symbols has one meaning. These systems are incapable of expressing irony, sarcasm, humor, or indirectness. Not so with language. A single ASL sentence can function as a request for information, a command, or a statement. In any language, a single symbol or group of symbols may have different functions and, conversely, a single function may be realized by different symbols. For example, the simple question in ASL,

signs have different meanings? and What part of the sign is responsible for the difference in meaning?

Nonmanual signals are the fifth basic part of signs. Many signs in ASL require a nonmanual signal in order to be produced correctly. Nonmanual signals are the facial expressions that accompany certain signs. For example, the sign NOT-YET is usually made with the mouth open and the tongue slightly out; the sign FINISH is made with the lips protruded. Without these nonmanual signals, the signs are not correct. As we will see, nonmanual signals are important at all levels of the language: for individual signs (lexical level), for building new signs and for making sentences (morphological and syntactic levels), and in the use of language by signers (discourse level).

When analyzing the distinct parts of signs, it is helpful to remember the following three points:

1. Make sure that the parts are indeed the same and not just similar. For example, the handshape of RESPONSIBILITY is a Bent B, the same as the handshape in COMPARE, but only similar to the handshape in BOOK.
2. Sometimes two English words are represented by the same sign. For example, SHOULD may sometimes be glossed as NEED, but the form of the sign is identical. The movement may differ depending on the meaning.
3. There are items that look like ASL signs in that they have handshape, movement, location, and orientation, but neither their meaning nor their function is ASL. For example, the sign BECAUSE has the same movement as FORGET or the same location as SUMMER, but BECAUSE is not an ASL sign. It is the result of codes invented to represent English manually.

TOOLS FOR WRITING ASL

Linguists have different tools that they can use for writing ASL. One of these tools is *notation*, which allows linguists to write down signs and the nonmanual signals that accompany them. This can be done through two methods—*glossing* and *transcription*. The other tool is *translation*, which is the conversion of a message from one language into the grammatical form of another language.

Notation Methods

Glossing. A *gloss* is an English word that represents a sign. Linguists use glossing to write a signed utterance. Glossing is not the same as translating, but, like translating, it is sometimes a difficult task. A gloss of a signed story will be a series of English words, written in small capital letters, that correspond to the signs in the ASL story. Parts of English, such as plural markers, past-tense markers, and prepositions, do not appear in glossing unless they are produced in the specific story. The nonmanual features are indicated on a line above the sign glosses. Some basic conventions used for glossing are as follows:

1. Signs are represented with small capital letters in English; for example, CAT, HOUSE, STUDENT.

2. Lexicalized fingerspelled words are written in small capital letters and preceded by the # symbol; for example, #DO.
3. Full fingerspelling is represented by dashes between small capital letters; for example, M-A-R-Y.
4. Nonmanual signals and eye gaze can be represented on a line above the sign glosses or by placing the nonmanual signal after angle brackets around the part of the sentence that co-occurs with the nonmanual signal. For example, the English sentence *I love linguistics* can be glossed as follows:

t

 LINGUISTICS, PRO-1 KISS-FIST or <LINGUISTICS>t PRO-1 KISS-FIST

In this example, the line above the gloss and the brackets represent nonmanual signals. The *t* over the line and after the brackets represents topicalization, which is marked by raised eyebrows (topicalization and other nonmanual signals will be explained in unit 15). Both ways of representing nonmanual signals will be used in this textbook.

Transcription. In addition to glossing signs into English, linguists have seen the need to devise a system for describing the structure of signs. We will discuss two systems that have been developed for describing the handshapes, locations, and movements of signs—the Stokoe system and the Liddell and Johnson system. These systems will be explained in later units; however, as an introduction to the concept of transcription, it is important to know the following three points:

1. In order for linguists to transcribe the structure of signs, they need to agree on the symbols used for describing the signs. These agreed-upon symbols are known as *conventions*, and they provide linguists with a consistent and predictable tool for transcription.
2. It is important that the transcription labels used be as precise as possible. The particular label for a handshape, a movement, or a location, and the arrangement of the labels in a particular way reveal something about the structure of signs.
3. The system chosen for labelling the parts of signs is a direct reflection of the researcher's perspective on the structure of signs. Transcription conventions do not exist in a vacuum, independent of linguistic theory. This point will be returned to in detail in discussion of the Stokoe system and the Liddell and Johnson system.

Translation

Translation is the representation of a message in one language into another language. Written messages in Italian can be translated into written English, for example. Messages signed in ASL can also be translated into written English. However, unlike glossing, translations do not include the nonmanual markers and other structural features that are represented in glosses.

On the DVD, you will find a story called “The Snowmobile.” The gloss of the first sentence in this story is PRO-1 ONE STORY NEVER FORGET (PRO-1 is the first-person pronoun, “I”). This glossing provides an English word for every sign in the order the signs are produced (see Figure 11). You can see that even though these



Figure 11. An example of a glossed sentence.

are English words, this is not an acceptable English sentence, but it helps us see what the signs are. Each one of these signs can be analyzed for its handshape, location, and other characteristics with specially devised symbols, or notations, which we will see in the discussion of the Stokoe and Liddell and Johnson systems. Finally, we can translate this ASL sentence to an acceptable English sentence: “There’s one story I will never forget.” All three of the writing tools are helpful for studying sign language structure.

SUPPLEMENTAL READING

“Signs Have Parts: A Simple Idea,” by Robbin Battison (1980); pp. 242–253

Homework Assignment 1

1. For each sign listed below, find another sign that has the same parameters for handshape, movement, and location.

	<u>Same Handshape</u>	<u>Same Movement</u>	<u>Same Location</u>
Example: FEEL	SICK	HAPPY	PRO-1. poss "my"

- a. RESPONSIBILITY
 - b. FORGET
 - c. CUTE
 - d. ENJOY
 - e. BICYCLE
 - f. UGLY
 - g. BEST
 - h. WORSE
 - i. MONKEY
 - j. DISCUSS
2. What is the difference between the signs in each pair?
- a. SUMMER/DRY
 - b. RED/CUTE
 - c. SHORT/TRAIN
 - d. SIT/CHAIR
3. What does each pair of signs have in common?
- a. SUMMER/DRY
 - b. RED/CUTE
 - c. SHORT/TRAIN
 - d. SIT/CHAIR
4. List four signs that must have a nonmanual signal with them.
5. Look at "The Snowmobile" on the DVD. Gloss the first 90 seconds of the story. The first part is glossed here as an example.
- | | |
|---|---------------------|
| t | head nod, eye blink |
|---|---------------------|
- PRO-1 ONE STORY NEVER FORGET. HAPPEN PRO-1 ABOUT OLD TWELVE.
- PRO-1. poss FAMILY HAVE TWO HOUSE. ONE HOUSE NORTH #MICH THAT-AREA,
- MOTHER-FATHER HOUSE, AUNT UNCLE HOUSE HOUSES-IN-A-ROW.

UNIT

4

The Stokoe System

In unit 3, we saw that ASL signs have internal structure; that is, that they can be broken down into smaller parts. Those parts include handshape, location, movement, palm orientation, and nonmanual signals. In this unit, we examine the first system devised for the formal description of signs. Homework Assignment 2 introduced you to labelling systems. In the first section, you described the handshape for each sign; in the second and third sections you described the location and the movement of signs. You have probably discovered that there may be different solutions to the same problem. For example, the handshape of PREACH can be described as a 9 or as an *F*; there are different signs for LOBSTER; both BUSY and COMMUTE can be described as having a back-and-forth movement.

The realization that emerges from the homework assignment is that there is a need for consistency and uniformity in a descriptive system. Arguments can be made for choosing either 9 or *F* as the label for the handshape in PREACH, but once a choice of label has been made, it must be used consistently. In addition to consistency, there is a need for precision, so that if the movement in both BUSY and COMMUTE can be described as *back and forth*, some way must be created to uniquely describe the movement in each sign. The movement in some signs, such as APPROACH or DIVIDE, may be difficult to describe, making the need for precision in descriptions even more important.

THE STOKOE SYSTEM

William C. Stokoe devised the first system for describing signs. Before Stokoe, signs were thought of as unanalyzable wholes, with no internal structure. Stokoe was the first to suggest that signs could be analyzed in the same way that the units of spoken language can be analyzed. In 1960, Stokoe proposed that signs have three parts (parameters) that combine simultaneously. The three parts are the location of the sign, which he called the *tabula* or *tab*; the handshape, which he called the *designator* or *dez*; and the movement, which he called the *signation* or *sig*. Palm orientation and nonmanual signals were dealt with indirectly in the Stokoe system.

Stokoe referred to the three parameters as *cheremes*, from the Greek word *cheir*, for hand. He saw cheremes as meaningless elements that combine to form all signs, in the same way that phonemes combine to form words in spoken languages. Each parameter has a set of members known as primes. For example, handshape primes include A, B, and 5; location primes include face, nose, and trunk; movement primes include upward movement, downward movement, and movement away from the signer. Figures 12 and 13 show the symbols used for writing the signs of ASL, as they appear in *The Dictionary of American Sign Language* (1965) by William C. Stokoe, Dorothy C. Casterline, and Carl G. Croneberg. In Stokoe's system, cheremes were written down in a specific order—TD^S. That is, the location of the sign (tab) was written first, followed by the handshape (dez), and then the movement (sig). For example, the sign IDEA is written as follows:

∩ | ^

∩ indicates the forehead location, | represents the handshape, and ^ represents the upward movement. Another example is the sign for “I” or “me.” The midchest location is indicated by []. The handshape is represented by G, x represents contact, and ⊤ represents movement toward the signer. All together, it looks like this:

[] G_⊤^x

tab dez sig

Stokoe's system allows for some variations on the basic TD^S representation of signs. Signs with two hands are represented as TDD^S (for example, WITH ØA'A^x, where the x indicates the two hands contacting each other); signs with one movement and then another are shown as TD^{SS} (for example, MILLION B_aB_b^{x⊥x}, where x^{⊥x} represents the repeated contact on the palm of the weak hand).

In the Stokoe notation system, the weak hand is regarded as the location for the sign, as shown in the following example (see Figure 14).

COOK B_aB_b^{xax}

PICK-ON G_∧X_∨^x

The signs PRO-1.poss (“my”), PRO-1 (“I, me”), and THINK are similar but show some contrasts. Here is how they look in the Stokoe system (see Figure 15):

PRO-1.poss PRO-1 THINK

[] B_⊤^x [] G_⊤^x ∩ G_⊤^x

Tab symbols

1. Ø zero, the neutral place where the hands move, in contrast with all places below
2. □ face or whole head
3. ∩ forehead or brow, upper face
4. △ mid-face, the eye and nose region
5. ∪ chin, lower face
6. 3 cheek, temple, ear, side-face
7. II neck
8. [] trunk, body from shoulders to hips
9. \ upper arm
10. √ elbow, forearm
11. α wrist, arm in supinated position (on its back)
12. D wrist, arm in pronated position (face down)

Dez symbols, some also used as tab

13. A compact hand, fist; may be like 'a', 's', or 't' of manual alphabet
14. B flat hand
15. 5 spread hand; fingers and thumb spread like '5' of manual numeration
16. C curved hand; may be like 'c' or more open
17. E contracted hand; like 'e' or more claw-like
18. F "three-ring" hand; from spread hand, thumb and index finger touch or cross
19. G index hand; like 'g' or sometimes like 'd'; index finger points from fist
20. H index and second finger, side by side, extended
21. I "pinky" hand; little finger extended from compact hand
22. K like G except that thumb touches middle phalanx of second finger; like 'k' and 'p' of manual alphabet
23. L angle hand; thumb, index finger in right angle, other fingers usually bent into palm
24. 3 "cock" hand; thumb and first two fingers spread, like '3' of manual numeration
25. O tapered hand; fingers curved and squeezed together over thumb; may be like 'o' of manual alphabet

26. R "warding off" hand; second finger crossed over index finger, like 'r' of manual alphabet
27. V "victory" hand; index and second fingers extended and spread apart
28. W three-finger hand; thumb and little finger touch, others extended spread
29. X hook hand; index finger bent in hook from fist, thumb tip may touch fingertip
30. Y "horns" hand; thumb and little finger spread out extended from fist; or index finger and little finger extended, parallel
31. B (allochric variant of Y); second finger bent in from spread hand, thumb may touch fingertip

Sig symbols

32. ^ upward movement
33. v downward movement
34. ^ up-and-down movement
35. > rightward movement
36. < leftward movement
37. z side to side movement
38. τ movement toward signer
39. ⊥ movement away from signer
40. ± to-and-fro movement
41. α supinating rotation (palm up)
42. D pronating rotation (palm down)
43. ω twisting movement
44. η nodding or bending action
45. □ opening action (final dez configuration shown in brackets)
46. # closing action (final dez configuration shown in brackets)
47. 2 wiggling action of fingers
48. @ circular action
49.)(convergent action, approach
50. × contactual action, touch
51. ⅋ linking action, grasp
52. + crossing action
53. ⊙ entering action
54. + divergent action, separate
55. " interchanging action

Figure 12. Stokoe's symbols for writing the signs of American Sign Language.

Note: Reprinted by permission of the publisher, from W. C. Stokoe, D. C. Casterline, and C. G. Croneberg, *A Dictionary of American Sign Language* (rev.). (1976): x–xii, Silver Spring, MD: Linstok Press.

$\cup R^{\times} \text{ — } \times > \times$

(initial dez; tips of dez fingers touch lips or chin) $\cup$ restaurant. In some localities sign may be used for 'doughnut'.

$\cup R_{\perp}^{\times}$

(imit.; knuckles of dez touch tab so that fingers project outward)

$\cup$ cigar.

$\cup V_T^{\circ}$

$\cup$ read lips; $\cup$ speech reading, lipreading, oralist; $\times$ oral. May also be extended 'speech' and 'the organs of speech'.

See also $\Pi V_T^{\wedge}$.

$\cup \ddot{V}_o^{\circ}$

(imit.: fangs; may also be made in high zero-tab with or without left G-hand touching dez elbow) $\cup$ snake, serpent. See also synonym: $\underline{B}_o \sqrt{G}_{\perp}^{\circ}$.

$\cup V^{\times}$

(initial dez; index fingertip of dez touches chin) $\cup$ vinegar.

$\cup V_{\#T}^{\times \square \wedge}$

(imit.; dez touches chin, moves up and snaps open to full V with or without touching forehead) $\cup$ goat. In some regions used for 'cheese'. 'Goat' is also signed $\cup A^{\pi} 5_{\omega}^{\times}$.

$\cup W^{\times} \text{ — } \times \perp^z$

(initial dez) $\cup$ water.

This sign serves as first element in several compounds:

— :: $\emptyset C_o C_o^{\vee}$ 'rain'

— :: $\emptyset 5_o \circ 5_o^{\perp}$ 'river'

— :: $\emptyset B B_{\perp}^z$ 'stream'

— :: $\emptyset \sqrt{5_o \circ} / 5_o^{\sim}$ 'ocean'

— :: $V_o V_o^{\times} :: \emptyset \sqrt{5_o \circ} / 5_o^{\sim}$ 'salt sea'

All these are imitative of the flowing, meandering, or undulating nature of the referent. However, these signs are seldom used except for such uses as signing a poem when nonce compounds too are acceptable, e.g. 'water' plus 'quiet' for 'pond'.

Usually signers spell the names of lakes, beaches, rivers, and oceans. On the east coast a-c 'Atlantic City' and o-c 'Ocean City'.

Figure 13. Examples of Stokoe's transcription system.

Note: Reprinted by permission of the publisher, from W. C. Stokoe, D. C. Casterline, and C. G. Croneberg, *A Dictionary of American Sign Language* (rev.). (1976): 168, Silver Spring, MD: Linstok Press.



Figure 14. Weak hand as location.



Figure 15. Sign contrasts.

SUPPLEMENTAL READING

"Introduction," to *A Dictionary of American Sign Language*, by William C. Stokoe (1965; 1976); pp. 254–269

Homework Assignment 3

1. Using the Stokoe symbols for movement, location, and handshape, transcribe the following signs:

- | | |
|---------------------|-------------------|
| a. ENJOY | e. SUNDAY |
| b. BEGIN | f. EVERY SATURDAY |
| c. BROKE (NO MONEY) | g. KNOW |
| d. BUSY | h. NOT |

Building New Signs

In part 2, we defined phonology as the study of the smallest *contrastive* parts of language. In American Sign Language, signs are made up of hold segments and movement segments. A hold segment has handshape, location, orientation, and nonmanual features, and likewise, a movement segment has handshape, location, orientation, and nonmanual features.

In comparing the signs LOUSY, AWKWARD, and PREACH with the signs THREE-WEEKS, **THREE-DOLLARS**, and NINE-MONTHS, we saw that the handshape in LOUSY, AWKWARD, and PREACH has no separate meaning. In those three signs, as in many others, the handshape, location, orientation and nonmanual information combine to produce one meaning. The separate parts can be identified, but they do not each have separate meaning. That is not the case, however, in THREE-WEEKS, THREE-DOLLARS, and NINE-MONTHS. In these three signs, the handshape does have a separate meaning—it indicates a specific quantity. To change the handshape in the sign THREE-DOLLARS immediately changes the meaning of the quantity of money being signed. To change the handshape in THREE-WEEKS or NINE-MONTHS changes the meaning of the number of weeks or months being talked about. (This process will be discussed more in unit 13, when we talk about numeral incorporation.)

Phonology is the study of the smallest contrastive parts of language. The parts of language that we study in phonology do not have meaning. So when we study phonology and we look at the sign THREE-WEEKS, we are simply interested in the fact that the sign has a handshape, a location, an orientation, and a movement. The fact that the handshape has the specific meaning of the quantity three is part of morphology.

MORPHOLOGY

Morphology is the study of the smallest *meaningful* units in language and of how those meaningful units are used to build new words or signs. Put another way, morphology is the study of word formation, of how a language uses smaller units to build larger units.

UNIT 11

Compounds

COMPOUNDS IN ASL

In unit 10, we saw that one way that ASL can create new signs is by deriving nouns from verbs. In this unit, we will look at another way that ASL can create new signs. Sometimes a language creates new words by taking two words (free morphemes) that it already has and putting them together. This process is called *compounding*. Both English and ASL have many compounds. We will first look at some examples from English.

In English, the word *green* is combined with the word *house* to make the word *greenhouse*. The word *black* is combined with the word *board* to make the word *blackboard*. Some other examples of English compounds are *hattrack*, *railroad*, *bookcase*, *blackberry*, *showroom*, and *homework*.

When nouns are derived from verbs in English or in ASL, a regular pattern can be described. A pattern can also be described for the formation of compounds. In English, when two words come together to form a compound, two fairly predictable changes take place.

1. The stress (that is, the emphasis) is usually on the first word of the compound, and the stress on the second word is usually reduced or lost. When the word *green* and the word *house* come together to form the compound *greenhouse*, the stress is on the word *green*: *gréenhouse*.
2. A new meaning is created when two words come together to form a compound. For example, *greenhouse* does not mean a house that is green; it has the specific meaning of a place where plants are grown. *Blackboard* does not mean a board that is black; it means a board that is used for instructional purposes, which may be black, green, or brown.

The research done by Supalla and Newport on nouns and verbs in ASL has already been mentioned. Another researcher, Scott Liddell, has done a great deal of research on compounds in ASL (see Table 4 for examples of ASL compounds). He noticed that when two signs come together to form a compound, predictable

Table 4. ASL Compounds

ASL Compound	English Translation
GIRL SAME	"sister"
BOY SAME	"brother"
MOTHER FATHER	"parents"
BLUE SPOT	"bruise"
THINK MARRY	"believe"
THINK SAME	"it's like"; "for example"
THINK TOUCH	"be obsessed with"
TALK NAME	"mention"
FACE NEW	"stranger"
GOOD ENOUGH	"just barely adequate"
JESUS BOOK	"Bible"
LOOK STRONG	"resemble"

Note: The symbol between the two glosses indicates that the sign is a compound.

changes take place as the result of rule application, just as they do in English-compound formation. There are two kinds of rules that cause the changes—morphological and phonological.

Morphological rules are applied specifically to create new meaningful units (in this case, compounds). Three morphological rules are used to create compounds in ASL: (1) the first contact hold rule, (2) the single sequence rule, and (3) the weak hand anticipation rule.

1. Sometimes the hold segment of a sign includes contact on the body or the other hand (+c). In compounding, the first or only contact hold is kept. This means that if two signs come together to form a compound and the first sign has a contact hold in it, that hold will stay. A preceding movement may be deleted, as may noncontact (–c) holds. If the first sign does not have a contact hold but the second sign does, that contact hold will stay. It is important to notice that while the hold may appear in the compound, the actual contact may not. For example, the sign GOOD has the structure

H M H
+contact –contact

The sign NIGHT has the structure:

X M H(+)
 +contact

The (+) following the H in NIGHT means that the sequence X M H is repeated. When the sign GOOD and the sign NIGHT come together to form the compound GOOD NIGHT, the first contact hold of GOOD is kept, and one movement-hold sequence of NIGHT is kept. The transition to the compound is as follows:

UNIT 12

Fingerspelling

FINGERSPELLING

ASL creates new signs in a third way—by representing the symbols of written English with ASL signs. This process is commonly referred to as fingerspelling. We will refer to these signs as fingerspelled signs.

Robbin Battison, an ASL linguist, did the first research on fingerspelling in ASL. He noticed, among many things, that when a written English word is represented with ASL signs, different changes may take place. It is important to notice that what have traditionally been called the “letters” of fingerspelling are ASL signs, each with a segmental structure and a handshape, location, and orientation. It is true that the handshapes of the signs may resemble the written symbol, and it is true that fingerspelling in ASL is the direct result of language contact with English. For example, the handshape of the sign C may look like the written English symbol C, but the sign is a sign and not a letter.

From a morphological perspective, these signs are free morphemes. A signer may produce each morpheme distinctly in what we will call full fingerspelling. **This is represented with dashes, as in W-H-A-T.** In actual production, however, changes often take place when fingerspelling morphemes are produced in sequence. A number of separate morphemes may begin to act like one single morpheme, like a single sign. This what we refer to as lexicalized fingerspelling, and we use the symbol # to mark it (see Table 5).

LEXICALIZED FINGERSPELLING

Many people have noticed that the separate signs of fingerspelling tend to blend together when they are produced in fingerspelled signs. That is, they tend to “become like individual signs.” In linguistics, the word *lexicalized* means “like a word,” or “word-like,” that is, like an independent unit. Examples of lexicalization in English include compounds such as *greenhouse*, *breakfast*, and *Christmas*, which are formed by uniting two separate lexical items that function as one word with a unique meaning. Acronyms such as NASA (National Air and Space Administration) and

Table 5. Lexicalized Fingerspelled Signs in ASL

#BANK	#DO
#BACK	#SO
#OFF	#OK
#ON	#KO
#IF	#JOB
#SALE	#YES
#EARLY	#NO
#BUT	#DOG
#BUS	#TOY
#CAR	#FIX
#HA	#WHAT

scuba (self-contained underwater breathing apparatus) are also examples of lexicalization in English. In these cases, a new word is formed by using the first letter of each word in the phrase.

Lexicalization describes the process of fingerspelling because the separate signs do seem to become like one, to be used like other ASL signs, and to follow the rules of ASL signs. For example, Battison noticed that in general, no sign uses more than two handshapes. This means that a sign like #IF or #OR can preserve both signs and still follow the rules of ASL. However, signs like #BACK or #EARLY present problems because they are formed from four and five signs. The result is that while all of the signs are not immediately lost, there is a tendency to reduce the number of signs as they become more like other ASL signs.

There is a difference between full, formal fingerspelling and lexicalized fingerspelling (see Figure 29), but it is easy to see how quickly the process of lexicalization begins. Just think about how you would fingerspell someone’s name if you were introducing them for the first time and then how the form of that fingerspelling would change if you used the name over and over again in a conversation. The changes that you observe are examples of lexicalization.

THE LEXICALIZATION PROCESS

Eight of the changes that are part of the lexicalization process are described in this section. These changes were first described by Battison (1978).

Some of the Signs May Be Deleted

In the lexicalization of #YES, there is a sign Y and a sign S; there is no sign E. While there are signs in ASL with one handshape or two handshapes in sequence, there are very few signs with more than two handshapes in sequence. However, many fingerspelled signs start out with four or more handshapes (for example, #BACK, #RARE, #SURE, #WHAT, and #EARLY). It seems that fingerspelled signs undergo pressure to conform to the rules of ASL structure. One of these rules seems to be “no more



Figure 30. Numeral incorporation in ASL.

the movement—and the location, orientation, and nonmanual signal. It means WEEK. The other meaningful part is the handshape, which has the meaning of a specific number. When the two parts are produced together, the meaning of the sign is “specific number of weeks.” A diagram of the two morphemes would look like this:

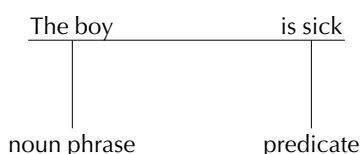
NUMBER OF WEEKS			
	H	M	H
Handshape (varies)			
Location	Base of hand		Tip of hand
Orientation	Palm down		Palm down
Nonmanual signal	—		—

The morphemes in this example are *bound* morphemes, that is, morphemes that must occur with other morphemes. For example, the handshape cannot occur by itself. It must occur within a segmental structure, with a location, an orientation, and possibly a nonmanual signal. Bound morphemes are different from *free* morphemes, which may occur by themselves. For example, the sign LOUSY in ASL (along with many other lexical signs) is a free morpheme. Its individual parts—handshape, location, orientation—do not have independent meaning and are not morphemes, but when they are all put together, the result is one meaningful unit, one morpheme. It is interesting to see the difference between the lexical sign LOUSY and the sign THREE-WEEKS. In LOUSY, the individual parts do not have independent meaning and are not morphemes, but the whole sign is a morpheme. The sign THREE-WEEKS has the same handshape as LOUSY, but in THREE-WEEKS, the handshape does have independent meaning and is a bound morpheme. In other words, the sign THREE-WEEKS is made up of two morphemes. What is interesting is that two signs with the same handshape can have such different linguistic structure.

The process of numeral incorporation is very common in ASL. Usually there is a limit to how high the numbers can go. For example, for most native signers, the handshape for WEEK can be changed from 1 through 9; for numbers 10 and higher

Basic Sentence Types

Languages have ways of referring to things or activities, and those ways are called *nouns* or *noun phrases*. Languages also have ways of saying something about those nouns or noun phrases, and those ways are called *predicates*. In the English sentence *The boy is home*, *the boy* is the noun phrase and *is home* says something about the boy. In this sentence, *is home* is the predicate. Predicates can have different forms; they are not limited to verbs. In fact, in the English sentence *The boy is home*, the predicate is a verb (*is*) plus a noun (*home*). In the English sentence *The boy is sick*, the predicate is a verb (*is*) plus an adjective (*sick*). We can look at the sentence like this:



Many languages do not use the verb *to be*. In those languages, a predicate may consist of simply a verb, a noun, or an adjective. In ASL, for example, the sentence BOY EAT consists of a noun (BOY) and a predicate, the verb EAT. The ASL sentence BOY HOME consists of a noun (BOY) and a predicate that is a noun, HOME. The ASL sentence BOY SICK consists of a noun and a predicate that is an adjective, SICK. These ASL sentences do not include the verb *is*, but the noun HOME and the adjective SICK function as predicates; they say something about the noun BOY. Verbs, nouns, and adjectives can be predicates in ASL.

In this unit, we will look at some basic sentence types in ASL, focusing specifically on the role that nonmanual signals play in ASL syntax. These six basic sentence types are questions, negations, commands, topicalization, conditionals, and declaratives.

Vocal intonation in English can mark different sentence types. For example, questions have a rising intonation at the end, while declarative statements have a falling intonation. In ASL, this kind of sentence-type marking is done using

nonmanual signals—movements of the face, head, and body that differentiate the different types of sentences and the clause units within them.

When we gloss ASL sentences, we need to be sure to include a written label for the grammatical nonmanual signals as well as the English words we have chosen to represent each sign. The typical way of representing nonmanuals is by writing the label on a line above the glosses with which the nonmanual co-occurs. An alternative method is to enclose the glosses in angle brackets and place the nonmanual label outside the right bracket. We will see examples of the first method throughout this unit as we discuss several of the sentence types used in ASL.

QUESTIONS

Yes-No Questions

Yes-no questions are questions that are answered with either a yes or a no; for example, *Is John home?* When a speaker of English asks a yes-no question, his voice usually rises at the end of the question. This specific vocal inflection is used in English to mark yes-no questions. Yes-no questions in ASL are marked using specific nonmanual signals. When someone asks a yes-no question in ASL, the eyebrows are raised, the eyes are widened, and the head and body may tilt forward; sometimes the shoulders are raised and sometimes the last sign is held (see Figure 31). To represent the nonmanual signal in writing, we draw a line over the signs that are produced at the same time as the face and body are producing the yes-no question nonmanual. The label that goes with the yes-no question nonmanual is *y/n*, so we write *y/n* at the end of that line. An example of a yes-no question is as follows:

 y/n
MAN HOME

This sentence would be translated into English as “Is the man home?”

Sometimes yes-no questions in ASL include a sign that is glossed as QUESTION-MARK. These questions are often asked when the signer is surprised by the information she is being given, or when the signer wants to check what the other person is saying. The QUESTION-MARK sign is made with an X handshape with internal move-



Figure 31. Nonmanual signals as part of a yes-no question.



Figure 32. An example of a yes-no question with the sign QUESTION-MARK.



Figure 33. An example of a Wh-question.

ment (wiggling). The resulting emphasis added by the QUESTION-MARK sign conveys unexpectedness. A similar English example would be the sentence *You really think that the test is easy?* (see Figure 32).

Wh-questions

Wh-questions involve the use of the words *who*, *what*, *where*, *when*, *why*, and *how*. Examples of Wh-questions in English are *Where is John?* and *When is class finished?* When a speaker asks a Wh-question, his voice usually goes down at the end of the question. This is the specific intonation that marks Wh-questions in English. Wh-questions in ASL also include the use of the signs WHO, WHAT, WHERE, WHEN, and WHY, and a very specific nonmanual signal. The nonmanual used to mark a Wh-question includes squinting the eyebrows and tilting the head; also, the body may lean slightly forward and the shoulders may be raised (see Figure 33). The label used for a Wh-question is *wh*, so we write *wh* at the end of the line marking the signs that include the nonmanuals. For example, the English sentence *Where is the man?* is glossed as follows:

wh
MAN WHERE



rh
PRO-1 TIRED WHY STUDY ALL-NIGHT

Figure 34. An example of an utterance containing a rhetorical question.

Rhetorical Questions

Sometimes a sentence contains a question that the asker does not really expect the other person to answer. These embedded questions are called *rhetorical questions*. The function of these structures is to provide the connections between related comments. They are called rhetorical questions because while they look like questions, they are not seeking a yes or no answer or any information from the other person. The signer provides both the question and its answer as part of her comment. The label that is used for rhetorical questions is *rh*, and an example of how it is used with a rhetorical question in ASL is as follows:

rh
PRO-1 TIRED WHY STUDY ALL-NIGHT

A literal translation of this sentence could be, “I’m tired. Why? Because I studied all night,” but the overall meaning of the ASL sentence is actually more parallel to the English sentence, “I’m tired because I studied all night.” The question *WHY* functions to connect the two comments, which is similar to the function of *because* and other conjunctions in English.

The nonmanual signal used with rhetorical questions is slightly different than the nonmanuals used with yes-no or Wh-questions. For rhetorical questions, the nonmanual includes raised eyebrows and a slight shake or tilt of the head (see Figure 34).

NEGATION

Sentences can be *affirmative* or *negative*. The process of changing an affirmative sentence to a negative sentence is called *negation*. A negative sentence states the opposite of the affirmative statement. For example,

affirmative: The man is home. negative: The man is not home.

In ASL, negative sentences have specific nonmanual signals that include shaking the head from side to side and, possibly, frowning or squinting. The symbol used for glossing negation in ASL is *neg*. An example, which could be translated to English as “The man is not home,” is as follows:

neg
MAN HOME



Figure 35. An example of a negated utterance.



Figure 36. An affirmative grammatical sentence with negative connotation.

The sign NOT is not required in negative sentences and may be used more for emphasis. The nonmanual signal is sufficient to produce a negative sentence (see Figure 35).

It is important to understand that grammatical negation does not mean that the words in the sentence are describing something bad or unwanted. Words that have negative connotations can be used in grammatically affirmative sentences and vice versa. For example, the sign DISGUSTING has a meaning that is understood as bad or negative, but the sentence shown in Figure 36 is grammatically affirmative.

COMMANDS

The sentences that people use to give commands are different from other kinds of sentences. In English, commands (or imperatives) often occur with vocal stress for emphasis, as in *Sit down!* or *Come here!* In ASL, imperatives have particular nonmanual signals, including making direct eye contact with the person being talked to and possibly frowning. An example of an imperative in ASL is *SIT*. The asterisk (*) is used before and after nonmanually marked signs to indicate an imperative.

TOPICALIZATION

Topicalization occurs in ASL when information is expressed at the beginning of the sentence to draw specific attention to it or to emphasize it. Often the topic names

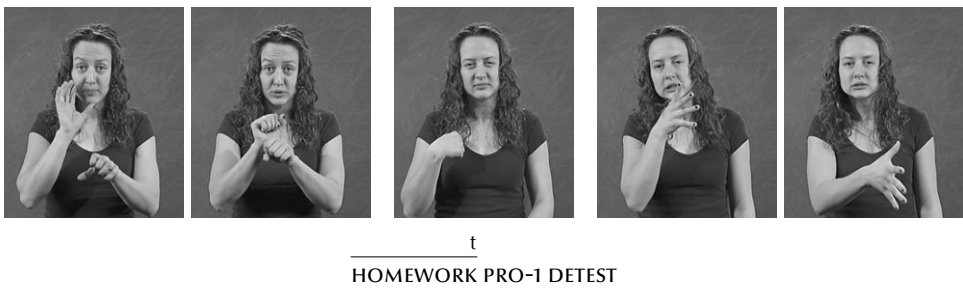


Figure 37. An example of topicalization.

what the rest of the sentence will be about. For example, in the ASL sentence PRO-1 DETEST HOMEWORK, PRO-1 is the subject, DETEST is the verb, and HOMEWORK is the object. If a signer wants to emphasize “homework,” she might produce the sentence with HOMEWORK nonmanually marked as a topic. The topicalized signs are marked by particular nonmanual signals, including raised eyebrows, a head tilt, and, possibly, a short pause. The label used for glossing the nonmanuals of topicalization is *t*, as shown in the following example.

t
HOMEWORK, PRO-1 DETEST

This could be translated literally into English as *Homework, I detest it*; it also could be translated, *As for homework, I really hate it* (see Figure 37).

CONDITIONALS

Sentences with conditionals express an if-then situation. In English, words such as *if* indicate a conditional, and the if-phrase can come at the beginning or end of the sentence. For example, in English you can say *If it rains tomorrow, the game will be cancelled*, or you can say *The game will be cancelled if it rains tomorrow*. In ASL, conditional phrases can begin with #IF and SUPPOSE, but they don’t have to include these signs; it is the use of conditional nonmanual signals that mark the phrase as conditional. These nonmanual signals include raised eyebrows, a head tilt, and possibly a short pause. If the nonmanuals are used, the phrase is conditional, with or without the use of particular signs. The symbol used for conditionals is *cond*, and the English sentence *If it rains tomorrow, the game will be cancelled* is glossed as follows:

cond
TOMORROW RAIN, GAME CANCEL

It is important to note that the second part of the sentence is not conditional. In ASL the conditional phrase is expressed first, and the statement that follows it may be negative or affirmative, a question, or a command (see Figure 38).



Figure 38. An example of a conditional clause in an utterance.

DECLARATIVES

Declarative sentences are statements that convey referential information. As a group, declarative sentences in ASL do not seem to be marked by a nonmanual signal, so when we gloss them, we do not include a line over the signs.

THE IMPORTANCE OF NONMANUAL SIGNALS

In ASL, nonmanual signals, and not the signs, determine the sentence type. The signs of a sentence can be identical, but the nonmanuals make the difference in sentence type. Consider, for example, the signs HOME PRO. These two signs can occur in four different sentence types. The signs themselves are the same; the nonmanuals are different. In each of the following examples, the nonmanual signal is vital in conveying the signer’s intended message.

Declarative:	HOME PRO	<i>You are home.</i>
	y/n	
Yes-no question:	HOME PRO	<i>Are you home?</i>
	neg	
Negation:	HOME PRO	<i>You aren’t home.</i>
Command:	*HOME PRO*	<i>Go home.</i>

SUMMARY

We have seen how different types of sentences are structured and how important nonmanual signals are in ASL syntax. Table 6 summarizes the information on sentence types and nonmanual signals in ASL.

Table 6. Sentence Types and Accompanying Nonmanual Signals

Sentence Type	Nonmanual Signals	Example
1. Questions		<u>y/n</u>
a. Yes-No Questions	Eyebrows raised, eyes widened, head and body may be tilted forward; shoulders may be raised; last sign may be held	MAN HOME
b. Wh-Questions	Eyebrows squinted, head tilted; body may be forward; shoulders may be raised	<u>wh</u> MAN WHERE
c. Rhetorical Questions	Eyebrows raised, head may be tilted or may shake slightly	<u>rh</u> PRO-1 TIRED WHY STUDY ALL-NIGHT
2. Negation	Head shakes side-to-side; may have frown or squint	<u>neg</u> MAN HOME
3. Commands	Direct eye contact with addressee, may frown	*SIT*
4. Topicalization	Eyebrows raised, head tilted, possibly a short pause	<u>t</u> HOMEWORK, PRO-1 DETEST
5. Conditionals	Eyebrows raised, head tilted; possibly a short pause and eye gaze shift	<u>cond</u> TOMORROW RAIN, GAME CANCEL
6. Declaratives	None	PRO-1 DETEST HOMEWORK

REFERENCE

Baker-Shenk, C., and Cokely, D. 1980. *American Sign Language: A teacher's resource text on grammar and culture*. Washington, DC: Gallaudet University Press.

UNIT 16

Lexical Categories

Now that you are familiar with the basic sentence types used in ASL, we can begin to explore additional aspects of ASL syntax. Before we do this, however, we need to take a look at how signs function in sentences. That is, do the signs function as nouns, verbs, adjectives, or adverbs? These different categories of signs are called *lexical categories*.

Large groups of lexical signs in ASL have very similar properties. These shared characteristics allow us to organize lexical signs into lexical categories. There are two types of signs in ASL. The signs that contribute to the substantive meaning of sentences are called *content signs*, and they include nouns (N), verbs (V), adjectives (Adj), and adverbs (Adv). The other type of signs are *function signs*, and they include pronouns (Pro), prepositions (Prep), and conjunctions (Conj). Content signs are included in the *open* lexical category because new signs can be added to them. In contrast, function signs are a *closed* lexical category because they do not allow new signs to be added.

Each lexical category has a unique set of morphological frames (the position of a sign with respect to the bound morphemes that can be attached to it with a sign) and syntactic frames (the position in which a sign occurs relative to other classes of signs in the same phrase). Both frames of a given sign can be used to determine the lexical category of that sign. The characteristics of the lexical categories in ASL are explained in the following section. Even though there are some universal tendencies across languages in the area of lexical categories, it is important to note that the lexical categories in ASL have unique characteristics that are not applicable to other languages. It is also important to point out that signs can be members of more than one lexical category. A sign's lexical category can only be identified when we see how it is used in a sentence.

OPEN LEXICAL CATEGORIES

Nouns

Noun signs identify entities such as individuals (name signs like DAVID-on-temple, #ANN), places (CHICAGO, #SEARS, #DENNY'S), and concrete and abstract things (COMPUTER, TABLE, THEORY).

Nouns have some common characteristics that can be used to identify them. One is whether or not they refer to things that can be counted. While many English nouns form the plural by adding the bound morpheme *-s* (*door/doors*), noun signs in ASL tend to occur only in the syntactic frame. That is, unlike English nouns, it seems that there are no bound morphemes that attach to nouns in ASL to pluralize or to otherwise modify them. A small number of ASL nouns form the plural by *reduplication*. That is, the noun is repeated two or three times in an arc or linear movement path. Some examples of noun reduplication are BROTHER++, SISTER++, WORD++, TREE++, and PLANT++. Most nouns, however, cannot be reduplicated. Another characteristic of nouns is that they can be modified with descriptive signs (adjectives). This will be explained later in the section on adjectives.

Verbs

Verb signs are actions, events, processes, and states of being: RUN, WALK, ENJOY, LIKE, HELP, INFORM. ASL has a dynamic verb system that will be explained in unit 19.

Adjectives

Adjectives are signs that describe nouns. Adjectives can describe an opinion, size, age, or color, among other qualities of a noun, and they usually appear before the noun. Figure 39 contains examples of adjectives paired with nouns.

Adjectives in ASL can also function like a verb or *adjectival predicate*. For instance in the sentence <MOVIE>t FUNNY, the adjective follows the noun and acts as a verb (see Figure 40). English adjectives can function in a similar way, except that they are used with a form of the verb *to be*, as in *That movie is funny*.

Adjectives can also be modified. The difference in meaning between TALL MAN and VERY-TALL MAN is that the first sentence describes a tall man and the second sentence describes a man who is very tall. The example on the left side of Figure 41 shows that the sign TALL is a two-handed sign in which the active hand moves from the base to the fingertips of the passive hand. When the sign begins well below the passive hand, brushes it, and ends above the fingertips, along with a marked facial expression, the sign can be glossed VERY-TALL (see right side of Figure 41).

Adverbs

Adverbs are signs that are used to express manner, indicate temporal frequency, or modify adjectives and other adverbs. English marks adverbs most often (although not always) by adding the suffix *-ly* to a word (*quick/quickly*). ASL signs are modified in a different way to express adverbial information. The articulation of a sign (how it

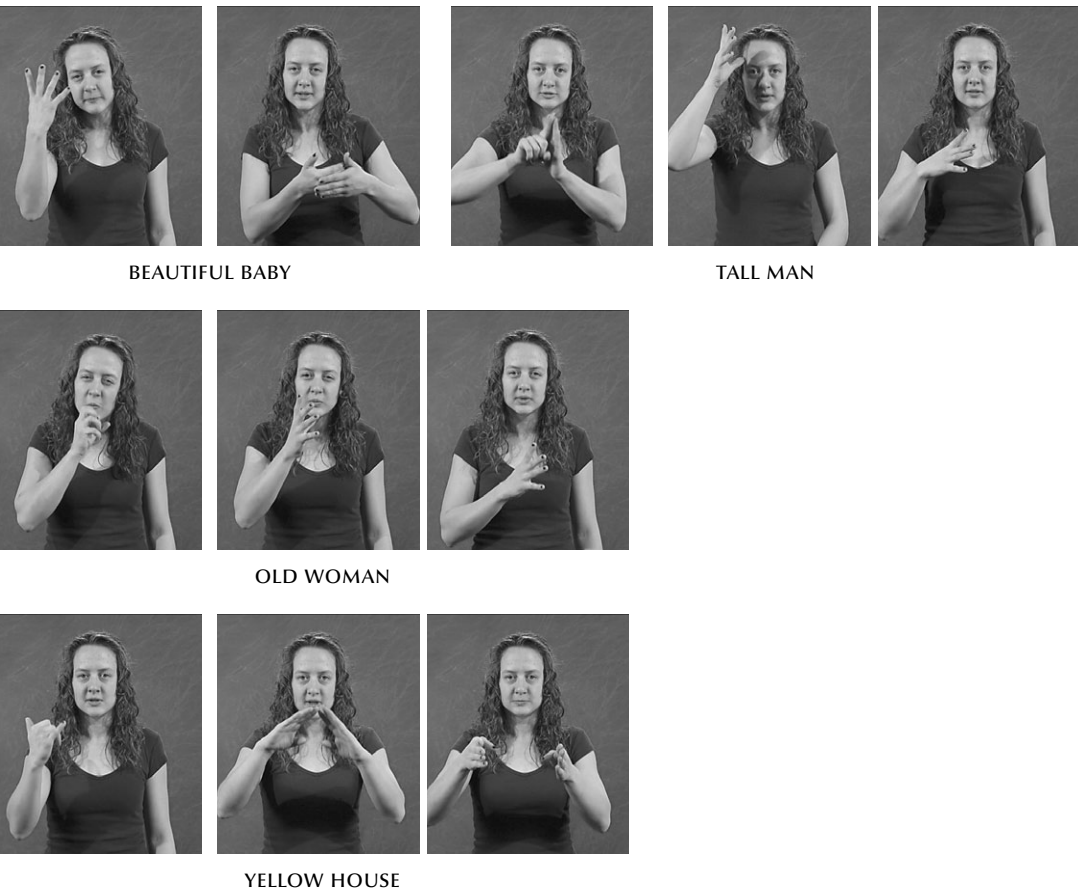


Figure 39. Examples of adjectives paired with nouns.



Figure 40. An ASL adjective functioning as a predicate.

is produced) can be altered and a nonmanual signal added, as in **<CHILD>t SIT-FOR-LONG-TIME** (see Figure 42). The sign **SIT** is produced with a repeated arc while the signer simultaneously puffs her cheeks. This sentence conveys the meaning that the child sat for an extended period of time.

Adverbs can also indicate when an action or event took place—**YESTERDAY**, **TWO-WEEKS-AGO**, **NEXT-TWO-DAYS**, and **STILL**. They tend to occur at the beginning of a sentence. Figure 43 shows a sentence with this syntactic frame.



TALL MAN



VERY-TALL MAN

Figure 41. An example of adverbial meaning incorporated into an adjective sign.



t
CHILD SIT-FOR-LONG-TIME

Figure 42. An example of altering the articulation of a sign accompanied by a nonmanual signal.

CLOSED LEXICAL CATEGORIES

The members of closed lexical categories often are used to relate various types of phrases to other phrases. They have little meaning outside of their grammatical purpose. These groups consist of pronouns (Pro), modal verbs (Mod), prepositions (Prep), and conjunctions (Conj).

Pronouns

A pronoun represents a person, place, or thing that has already been identified in a sentence. Languages vary in what information their pronoun systems encode (gender, number, person, case). Examples of pronouns in English are *he*, *she*, *it*, *them*, and *us*. Examples of English sentences with pronouns are *He came home early* and *She gave it to us*. When reading or hearing those sentences, we must know to what or to whom *he*, *she*, *it*, or *us* refers. If we do not know, we cannot understand the sentence. We understand the sentence because the referent (the noun that the pronoun represents) has been introduced earlier in the conversation or because we guess from the context. For example, if one sentence describes *the boy* and the next sentence uses the



Figure 43. An adverb indicating when an event will occur.



Figure 44. First- and nonfirst-person pronouns in ASL.

pronoun *he*, it is safe to assume that the pronoun *he* represents *the boy*. Or suppose three people are seated at a table. If one person looks at the person on her right and points to the person on her left and says *He told me something interesting*, the meaning of *he* comes from the context, and we can assume that *he* refers to the person on the left.

ASL also has pronouns, but its pronoun system differs from English in a number of ways. The first difference is that ASL only has first-person (PRO-1) and nonfirst-person pronouns (PRO) while English has first- (*I/me*), second- (*you*), and third-person (*he/him, she/her, it*) pronouns. For example, the ASL sentence PRO-1 PUNISH PRO has two pronouns, one that points toward the signer and usually contacts the chest (glossed as PRO-1), and one that points away from the signer (glossed as PRO). The English translation of this sentence could either be “I punished you” or “I punished him,” depending on where the signer was looking. Both sentences use the same ASL sign, PRO because there is not a distinct form separating second- and third-person pronouns (see Figure 44).

ASL and English also differ in how they distinguish between subject and object pronouns. English uses different words—*he* (subject pronoun) and *him* (object pronoun), *we* and *us*, *she* and *her*, *they* and *them*, and so forth. ASL pronouns, like the pronouns in many other languages, do not show this distinction. Instead, subject and object are indicated in the sequence of signs. For example, if the initial and final signs in Figure 45 were reversed, the meaning would change.

One similarity between ASL and English is that both languages distinguish between singular and plural pronouns. For example English speakers can say *I want a drink* or *we want a drink*, depending on how many people are thirsty. ASL makes the same distinction (PRO and PRO-PL). Compare the sentences in Figure 46.



Figure 45. Distinguishing between subject and object pronouns in ASL.

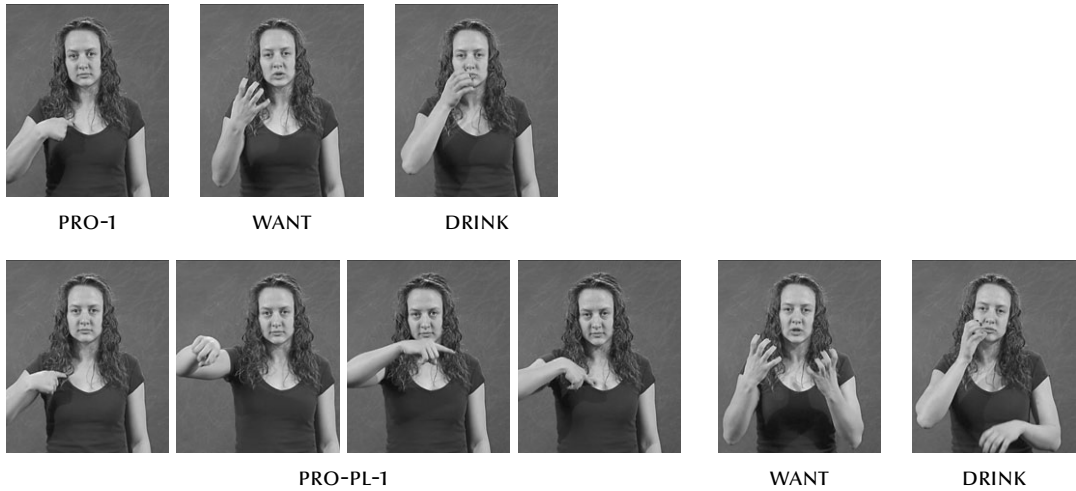


Figure 46. Distinguishing between singular and plural nouns.



Figure 47. Pronouns specifying number.

In addition, ASL has pronouns that are produced with numeral morphemes. This allows the pronouns to identify a specific number. These pronouns are produced with a horizontal circular motion, as can be seen in Figure 47. English can express the same concept that these ASL signs do, but it requires a phrase rather than a single pronoun, *the three of us* or *the four of us*. Table 7 summarizes the pronoun system described in this section.

Table 7. The Pronoun System in ASL.

First-person Examples				Nonfirst-Person Examples			
PRO-1				PRO			
PRO-DUAL-1				PRO-DUAL			
PRO-MULTI-THREE				PRO-MULTI-THREE			
PRO-MULTI-FOUR				PRO-MULTI-FOUR			
PRO-MULTI-FIVE				PRO-MULTI-FIVE			
PRO-PL-1				PRO-PL			



Figure 48. Examples of modals.

Modal Verbs

Modal verbs like *WILL*, *CAN*, *MUST*, and *SHOULD* express the idea of necessity or possibility. Modal verbs are produced with other verbs; the modal comes first, followed by the main verb. Occasionally the modal is also repeated after the verb for emphasis. Figure 48 illustrates some examples in the syntactical frame.

Emphasis can be added to modal verbs by incorporating nonmanual signals. For example, the difference between *should*, *need to*, and *absolutely must* can be conveyed by the intensity of the nonmanual signal. The last example in Figure 49 playfully shows intensity by the use of all four fingers in the sign *MUST*.

Prepositions

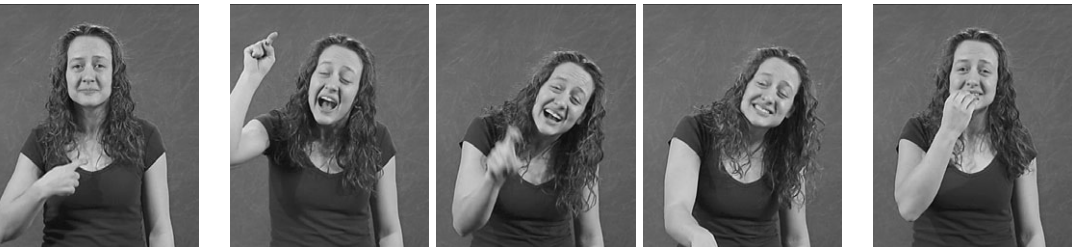
Prepositions (Prep) show relationships between nouns and predicates or pronouns. English has many words to express these relationships (e.g., *under*, *on*, *in*, *above*, *with*, and *to*). In ASL these relationships are typically expressed through depiction, indicating and depicting verbs, and the index finger pointing to mean “at.” Compared to English, ASL does not have many independent prepositions, but it does use some signs—*IN*, *UNDER*, and *BEHIND*. However, these signs have a different function in ASL in that they typically incorporate more information than English prepositions do. Signs including prepositional information often function as predicates. For example, when the sign *INSIDE* is produced on the chest with repeated



nodding
PRO-1 MUST EAT



nodding
PRO-1 MUST EAT



nodding
PRO-1 MUST EAT



nodding
PRO-1 MUST EAT

Figure 49. Use of nonmanual signals to intensify modals.

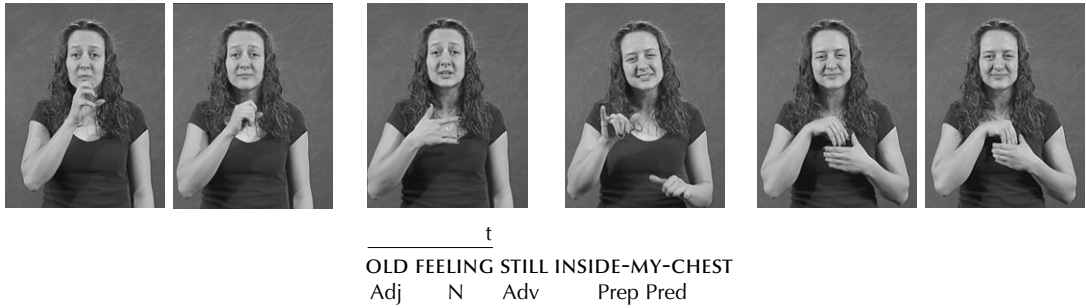


Figure 50. An example of prepositional predicate.



Figure 51. Examples of conjunctions.

movement to talk about inner feelings, it becomes the predicate of the sentence (see Figure 50).

Conjunctions

Conjunctions join words or phrases of the same category. ASL conjunctions include BUT, UNDERSTAND, #OR, and PLUS. Examples of conjunctions in the syntactic frame are shown in Figure 51.



Figure 52. Basic word order in ASL.

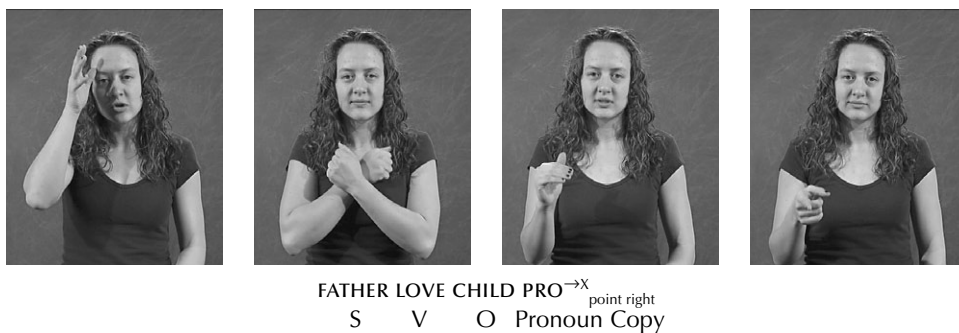


Figure 53. An example of subject pronoun copy.

For example, as shown in Figure 53, the pronoun copy PRO refers to the subject of the sentence, the father.

Topicalization

In Unit 15, we introduced sentences with topicalized components. Topicalization occurs when information is expressed at the beginning of the sentence to draw specific attention to it or set it up as prominent, and it results in an adjustment to the basic word order. Often the topic names what the rest of the sentence will be

about. The topic, or prominent information, is signed before the main clause of the sentence and is accompanied by a nonmanual signal. The nonmanual signal used in topicalization is raised eyebrows, slight forward head tilt, and a possible short pause. The use of topicalization allows for structures like the following to occur:

$$\begin{array}{ccc} \text{Topic} & \text{S} & \text{V} \\ \hline t & & \\ \text{CHILD, FATHER LOVE} & & \end{array}$$

The sign CHILD is the topic, and the signer makes a comment about the child. As we saw before, this same sentence can occur with pronouns instead of nouns, as follows:

$$\begin{array}{ccc} \text{Topic} & \text{S} & \text{V} \\ \hline t & & \\ \text{PRO}^{\rightarrow Y}_{\text{point right}} & \text{PRO}^{\rightarrow X}_{\text{point left}} & \text{LOVE} \end{array}$$

The direction where the signer points for PRO provides information about who is being referred to (see Figure 54).

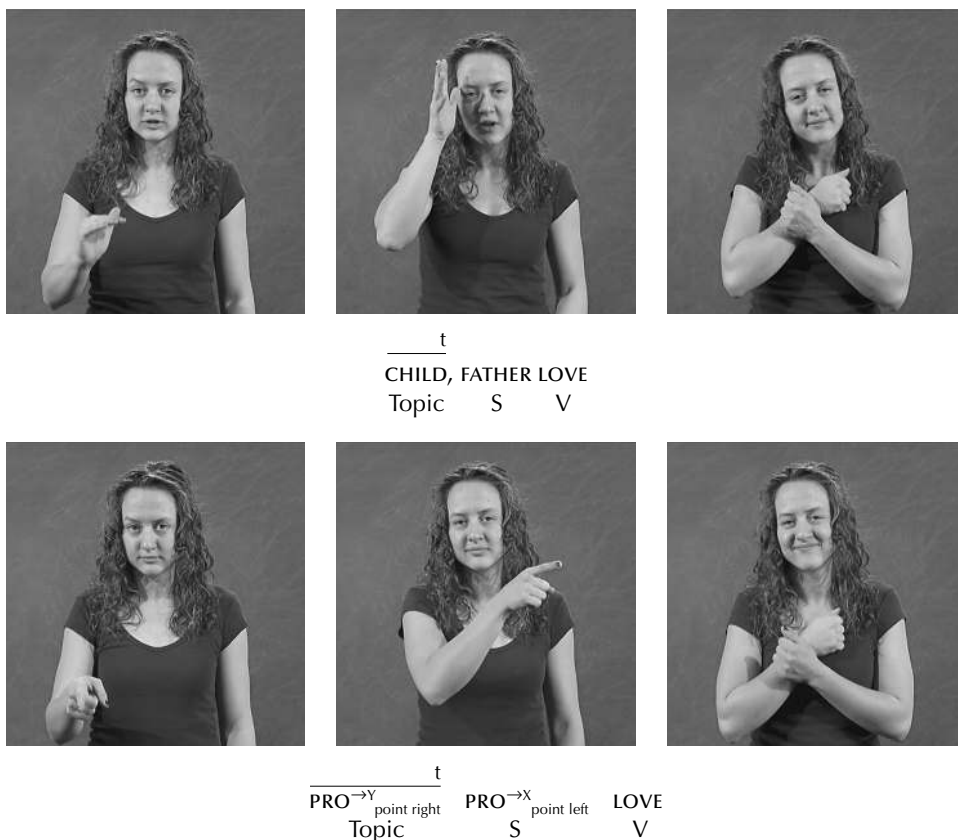


Figure 54. An example of topicalization.

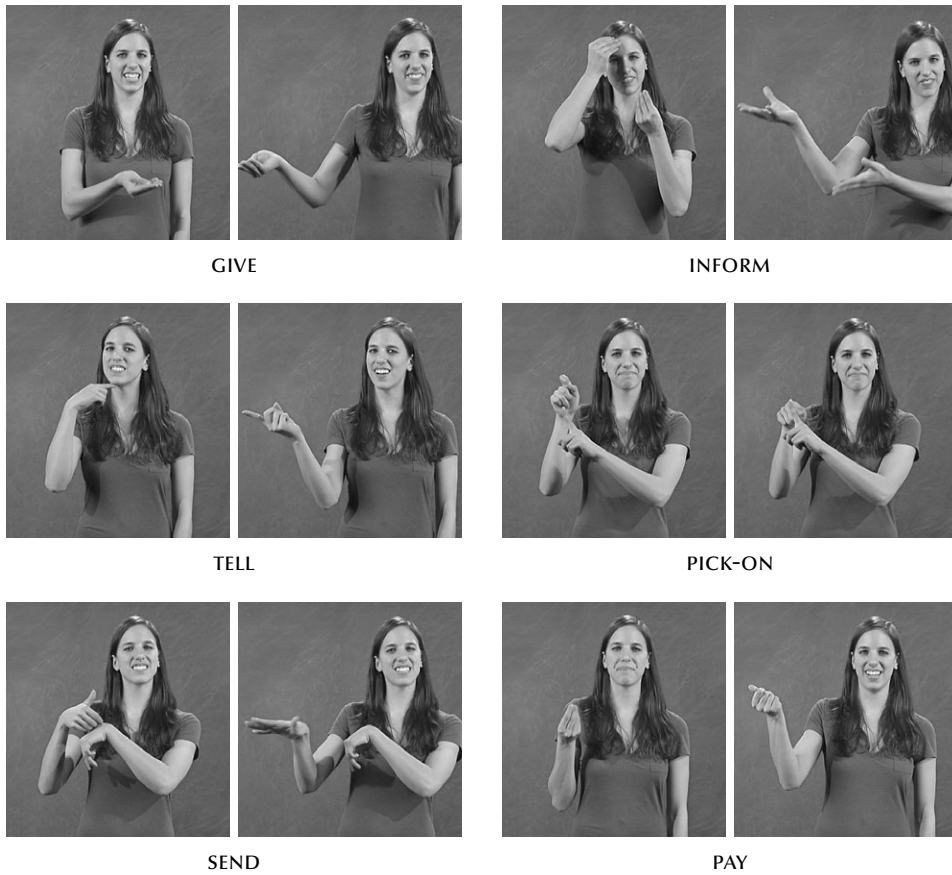


Figure 69. Examples of indicating verbs.



Figure 70. Subject and object information contained in the location of an indicating verb.

entity that is doing the detesting. In the ASL translation of the English sentence *I hate him*, the back of the hand faces the signer and the palm faces the location where the hated person has been established. Separate signs for *I* and *him* are unnecessary because the direction of the verb contributes this information. In the sentence *He hates me*, the situation is reversed (see Figure 71). For some verbs like



Figure 71. Subject and object information contained in the orientation of an indicating verb.



Figure 72. Subject and object information contained in the location and orientation of the indicating verb.

INVITE and HIRE, the first location of the verb is related to the object and the second location refers to the subject.

Orientation and location may be combined to contribute information about the subject and object. The ASL verb #SAY-NO-TO can be used to express the English sentence *I say no to him*. The signer would begin the sign near his or her body, with the palm facing the other person's location, and then the hand would move in the direction of the other person. To convey the meaning in the English sentence *He says no to me*, the orientation and beginning and ending locations of the verb #SAY-NO-TO are reversed. In this articulation, the sign would begin in the location of the other person and the palm of the hand would face the signer; the sign would end near the signer (compare the signs in Figure 72).

Indicating verbs can also add information by showing reciprocating action. These verbs are called *reciprocal verbs*. In a situation in which two people are looking at each other, they both are looking and being looked at simultaneously. The indicating verb LOOK-AT-EACH-OTHER captures this situation. Each hand represents a person. The fingers of the right hand point directly at the fingers of the left hand, thus showing how each person is surveying the other with their eyes. Another example of a reciprocal indicating verb is UNDERSTAND-EACH-OTHER. In this verb, one hand is placed near the signer's forehead, with the palm facing out. The location of the hand represents the signer, and the orientation points toward the person with whom the signer is conversing. The second hand is on the same level as the first

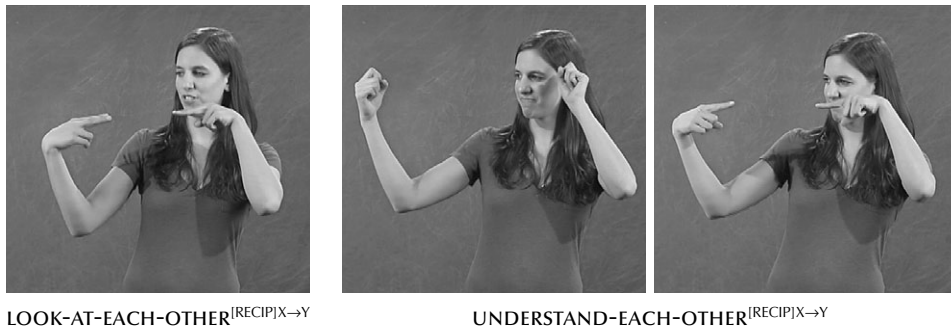


Figure 73. Two examples of reciprocal verbs.



Figure 74. The di3rection of locative verbs reveals specific information.

hand, with the palm facing the signer. The location of the second hand represents the other conversant. Each hand represents a person, and the fact that the palms are facing each other indicates that the signers comprehend what the other is saying (see Figure 73).

Locative verbs are another type of indicating verb. Locative verbs are different from plain verbs like ENJOY, PUNISH, or UPSET, in which the location feature is simply a part of how the sign is made. The function of the location in plain verbs is articulatory; the location does not have independent meaning. With locative verbs, the actual direction or location of the sign contains specific meaning. A good example of a locative verb is THROW. When a signer signs the sentence JOHN THROW[→] ROCK, the direction of the sign indicates the direction in which the object is thrown (in this case, away from the signer towards some location). The direction of the sign can vary, depending on the meaning the signer wants to convey. If the signer is talking about throwing something upward, like shooting a basketball, the direction of the sign is upward; if the signer is talking about throwing something into a trash can, the direction is downward (see Figure 74).

Other locative verbs include the sign HURT and the fingerspelled sign #HURT, in which the location shows the specific area of the body that hurts. While the location or direction of a locative verb represents a place in three-dimensional space, the handshapes of these signs do not have independent meaning. The best way to illustrate the difference is to compare the locative verb #HURT and the depicting verb USE-SCALPEL (see Figure 75). While we would say the handshape in USE-SCALPEL



Figure 75. The difference between locative and depicting verbs.



Figure 76. Depicting verbs showing the movement of a car.

is related to a hand holding a scalpel, the handshapes of #HURT do not contribute additional meaning.

DEPICTING VERBS

The final category of ASL verbs is *depicting verbs*. What distinguishes this category of verbs is their ability to convey two types of information. Depicting verbs, like other verbs, contain information related to action or state of being. Additionally, the forms of depicting verbs represent aspects of their meaning. In the past, many of the verbs in this category were labeled classifier predicates (see box). Examples of depicting verbs include VEHICLE-DRIVE-BY, PERSON-WALK, FLAT-SURFACE, VEHICLE-DRIVE-UPHILL. The examples in Figure 76 show the movements of a car. In the first sign, the car is being driven on a flat surface and in the second sign the car is being driven up a hill.

One way to describe how the form of depicting verbs represents aspects of their meaning is through *blending* (Liddell 2003a). The concept of blending is not limited to signed languages; it is part of the general way that people conceptualize and make sense of information. For example, if two people were sitting in an office having a conversation, and one wanted to describe the street where she lived, she might set up objects on the desk as a visual representation of her street. She might use a book to represent her house, a folder to represent her neighbors' house, and a pen to represent the railroad tracks at the end of the street. Each of these real items present on the desk would be understood to represent part of the imagined "scene" of the street.

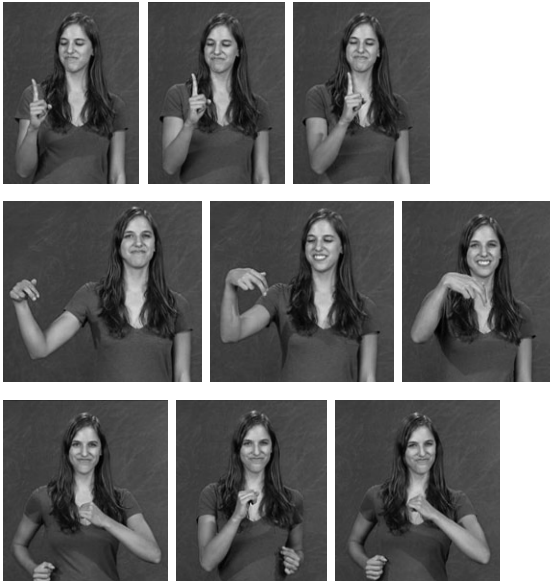


Figure 77. Three possible ways to depict a person walking.

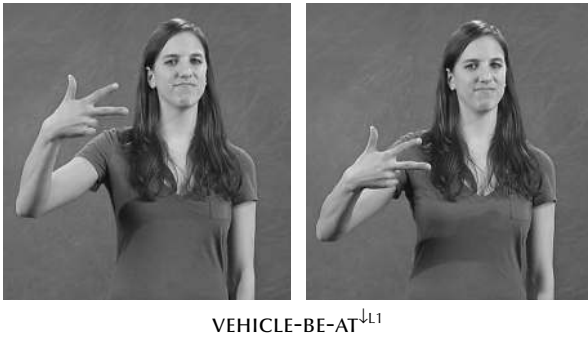


Figure 78. A depicting verb showing location in space.

is accomplished by using a depicting verb to locate the car in space. The verb is made with a short downward movement followed by a hold, and it is transcribed as VEHICLE-BE-AT^{↓L1} (see Figure 78). The superscript ^{↓L1} indicates that this depicting verb is placing an entity at a specific location using a downward movement.

The second type of depicting verb shows what something looks like, how it is shaped, or how objects are arranged. Returning to the story about the car ride, the narrator may want to describe the steep mountain that he traveled up. A depicting verb showing the mountain surface could be used. This type of depicting verb differs from the first in that one hand is kept stationary while the second moves away from it. The movement of the second hand shows the steepness of the hill the driver climbed, and it can be expressed as FLAT-SURFACE-EXTEND-UPWARD^{↓L1-L2}. The superscript ^{↓L1-L2} means that the sign moves from a starting location to an ending location. This type of verb is dynamic in that it can provide two types of informa-

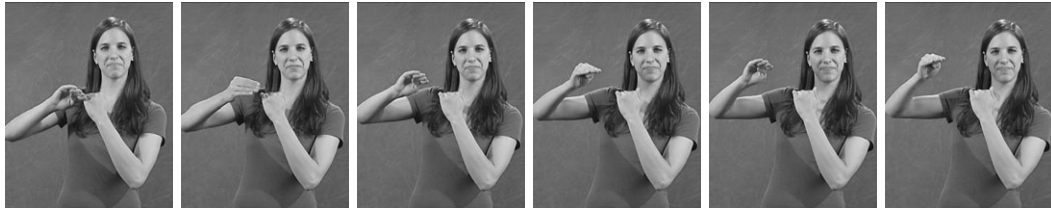
FLAT-SURFACE-EXTEND-UPWARD^{↓L1-L2}BUMPY-SURFACE-EXTEND-UPWARD^{↓L1-L2}

Figure 79. Depicting verbs can fully describe an object.

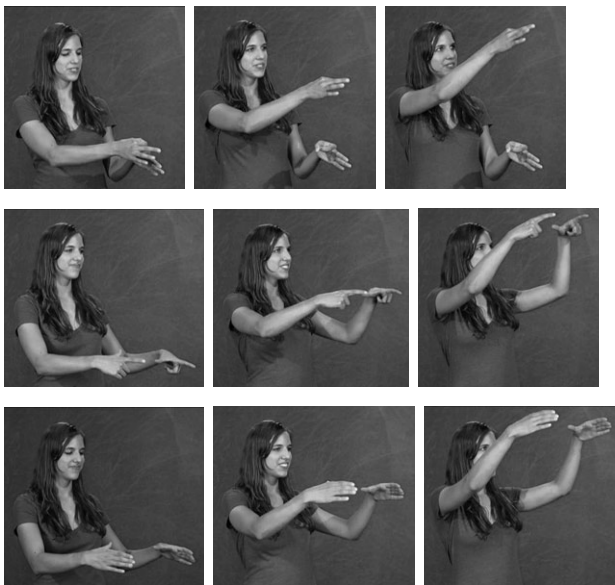


Figure 80. Depicting the size and shape of columns.

tion—what the surface looks like and where it extends. If the driver came to an unpaved portion of the road, the sign could be altered to reflect this (see Figure 79).

Signers also use this second type of depicting verb to describe how things look, including different sizes and shapes. The three examples in Figure 80 all depict columns. The different handshapes represent three different sizes of column, either actual different sizes or differences based on perspective.

The third type of depicting verb shows movement or action. Continuing the example of the narrator talking about his car, he would sign VEHICLE-DRIVE-TO^{↓L1-L2}

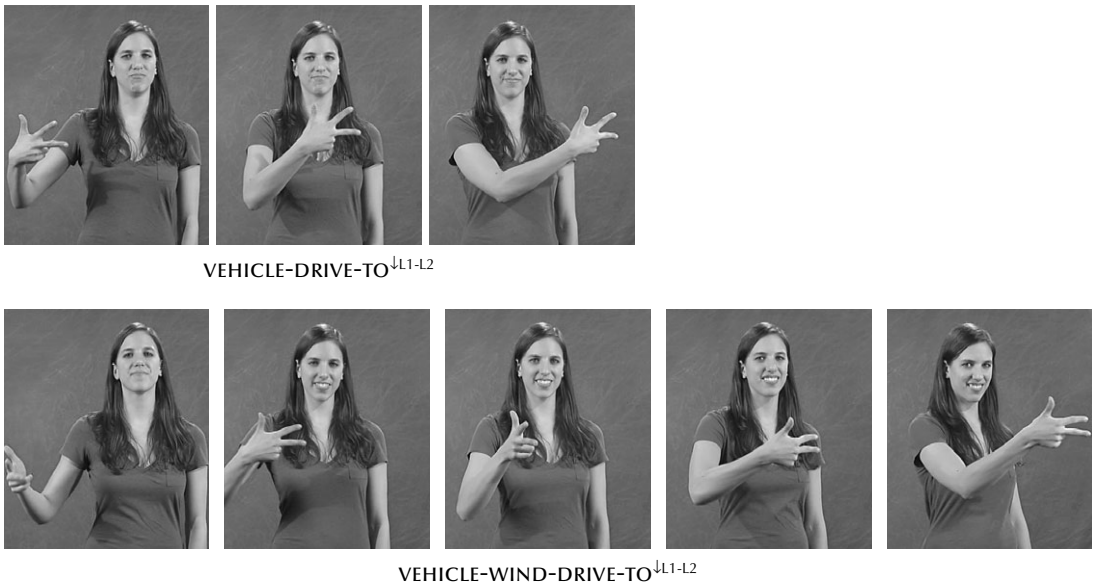


Figure 81. Depicting verbs showing the movement of a car.



Figure 82. Depicting a person drinking from a teacup.

to indicate that the car moved from point A to point B. As was the case with the second type of depicting verb, this third type can also reflect the path of the motion. If the car traveled on a winding road, the signer could express this as *VEHICLE-WIND-DRIVE-TO*^{↓L1-L2} (see Figure 81).

This category of depicting verbs, those that show action, includes depictions in which signers use the whole upper body—hands, arms, face, etc.—to represent people’s actions. The example discussed earlier of ways to depict a person walking (Figure 77) falls into this category. Other examples include signs depicting people holding and using objects, such as drinking from a teacup, as shown in Figure 82.

Some depicting verbs have been used widely and consistently over time in the Deaf community, to the point that they have become *lexicalized*. That is, they have assumed a meaning that is separate from the morphemes that make up the sign. They can function like plain verbs or as nouns in ASL utterances. For example, the sign *KEY* is clearly based on the depiction of a key being placed into a lock and



Figure 83. Depicting a person opening a door with a key.



<PRO-1.poss KEY WHERE>wh
Where is my key?

Figure 84. The noun KEY lexicalized from a depicting verb.

turned. One can express the action as a depicting verb, or the form can be used as a lexicalized noun in a sentence like <PRO-1.poss KEY WHERE>wh, meaning “Where is my key?” (See Figures 83 and 84.) Other examples of depicting verbs that have become lexicalized include DRINK, PACK, and FUNERAL. ASL also has some lexical signs that are understood based on metaphorical depiction, such as FALL-IN-LOVE and TRUST.

SUPPLEMENTAL READINGS

- “Indicating Verbs and Pronouns: Pointing Away from Agreement,” by Scott K. Liddell (2000); pp. 377–389
- “Body Partitioning and Real-space Blends,” by Paul G. Dudis (2004); pp. 390–400

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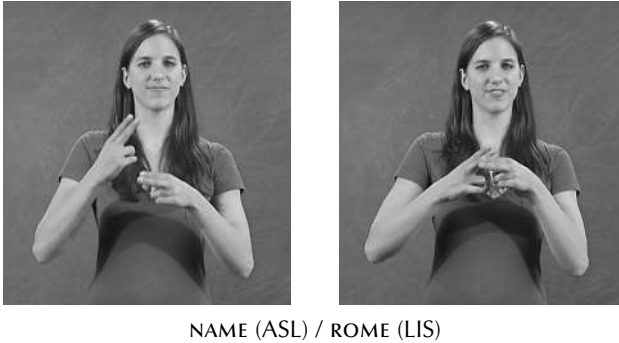


Figure 85. The same sign can have different meanings in different signing communities.

or signs by carefully observing the way words and signs are used by real people. Dictionaries for sign languages are unique in that, while they may be organized to reflect the linguistic structure of the sign language, they seem to be almost necessarily bilingual. That is, in addition to providing an inventory of the signs in a sign language, sign language dictionaries almost always also provide written glosses for the signs in the majority language of the community in which the sign language is used. *Stokoe's Dictionary of American Sign Language was an exception because it did not present the signs by an alphabetical list of their English glosses, but rather by the parameters of location, handshape, and movement.* The potential problem with bilingual dictionaries is that the same sign may have different English glosses assigned to it or different signs may be assigned the same English gloss. Therefore, the meaning of the sign may not always be clear from the gloss assigned to it.

Types of Meaning

Researchers have identified three different kinds of meaning: *referential* meaning, *social* meaning, and *affective* meaning.

Referential Meaning. Referential meaning is the idea, thing, or state of affairs described by the sign or sentence. For example, the sign CAT refers to a four-legged mammal with a tail, whiskers, and so forth. The meaning of the sign CALIFORNIA is the western state that has the Pacific Ocean to the west, Mexico to the south, Oregon to the north, and so forth. That state is the referent of the sign.

Social Meaning. Signs and sentences also provide information about the social identity of the language user. For example, certain choices of signs may reveal where the person is from, whether the person is male or female, or whether the person is African American or Caucasian. To illustrate this, imagine someone signing AWFUL with the strong hand pointing upward (Figure 86). Can you visualize a male signer using this sign? Most likely the answer is no. The reason for this is that most ASL users deem this production of the sign to be feminine. This is how social meaning is conveyed by this sign.

The Meaning of Sentences

WORD ORDER AND SEMANTIC ROLES

We have seen some of the ways in which individual signs have meaning. We now turn our attention briefly to the ways in which sentences have meaning. First of all, we will see that it is not enough to simply say that the meaning of a sentence comes from just adding up the meanings of all the signs in the sentence. There is more to it than that. We can see this from the following two ASL sentences:

CAT CHASE DOG *The cat chases the dog.*

DOG CHASE CAT *The dog chases the cat.*

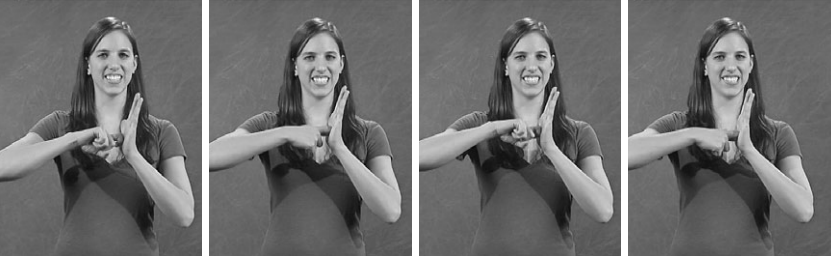
These two sentences have different meanings, and the difference in meaning comes from the *order* in which the signs appear. So, sign order or word order is an important factor in determining the meaning of a sentence. What is also important is the relative *semantic role* of each sign in a sentence. By *semantic role*, we mean how sentences show who did what to whom, with whom, or for whom. Examples of semantic roles are *agent*, *patient*, *experiencer*, *instrument*, and *cause*. For example, in the first sentence above, the cat is the agent, the “doer” of the action described by the verb, and the dog is the patient, the “receiver” of the cat’s action. In the second sentence, the same signs have opposite semantic roles—the cat is now the patient, and the dog is now the agent, showing us how important knowing the semantic role of a sign is in understanding a sentence. In the sentence JOAN LIKE PIZZA, Joan is the *experiencer*. She is not really doing anything or receiving any action; she is experiencing some physical or psychological sensation. In the English sentence *The key opens the door*, the semantic role of the word *key* is as *instrument*. The ASL version of that sentence might be as follows:

_____ t
DOOR, KEY, dv: KEY-OPEN-DOOR

It is interesting to note that the semantic role of instrument is often realized in ASL with the use of a specific handshape during depiction, along with the sign for the instrument (see Figure 91).



DOOR



KEY



KEY-IN-DOOR

TURN-KEY



KEY-OPEN-DOOR

t
DOOR, KEY, dv: KEY-OPEN-DOOR

Figure 91. An example of the semantic role *instrument*.